

METHODE ELHYDRO - CROP & RESOLUTION

1. What is this ?

The ElHydro method consists of cropping invisible or unnecessary peripheral pixels in VR (croptangential) to redirect all the saved GPU power into targeted oversampling where the eye is looking. Thanks to a dedicated mathematical calculator, it compensates for the reduced field of view to maintain perfect optical geometry while drastically increasing effective sharpness (PPD). The result is surgical visual clarity without resorting to destructive third-party drivers, all while locking in a maximum and stable framerate.

In simpler terms, the unnecessary edges of the image that you never look at while playing are cropped to reduce the load on your graphics card. The power freed up is then directly redirected to the center of your field of view, making the game much sharper while remaining perfectly smooth.

2. What is needed to implement this method?

- OpenXR Toolkit - PrimaShock mod (toolkit allows you to perform cropping operations, among other things)
- Excel file for his headset (the Excel file indicates the correct values that we will need later)

3. Install Openxr Toolkit - mod Primashock (version 1.4.0)

https://eu.pimax.com/blogs/blogs/a-guide-to-primashock-s-openxr-toolkit-mod?srltid=AfmBOoqR95Xx9pTTjZDHc7Bn20eE9YkWMZkvGw_X5-3CVLthO2JYGcYx&shpxid=0c032444-9a31-414b-8d5a-737d69f385d1

(Download links are also available at these links)

<https://github.com/mbucchia/OpenXR-Toolkit/releases/tag/1.3.2>

<https://primashock.com/openxr-toolkit-mod-by-primashock-vr/>

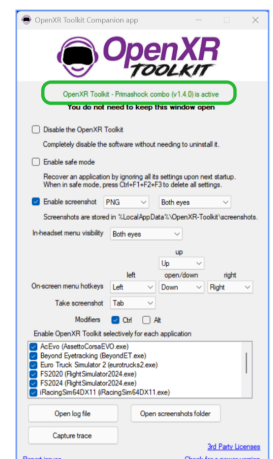
<https://discord.com/channels/1371247671024680990/1536426623157534750>

Step 1: Prerequisites Before installing the mod, make sure you have installed the original version 1.3.2 of OpenXR Toolkit.

1. Download the file [OpenXR-Toolkit-v1.3.2.msi](#).
2. Launch the installer and follow the instructions to complete the setup.

Step 2: Installing the PrimaShock mod Once the basic toolkit is installed, follow these steps to apply the modified files:

1. **Download the files:** Download the mod's DLL file via the ZIP file download button. Unzip it and extract the file(s).
2. **Locate the installation folder:** It is usually found underneath `C:\Program Files\OpenXR-Toolkit`.

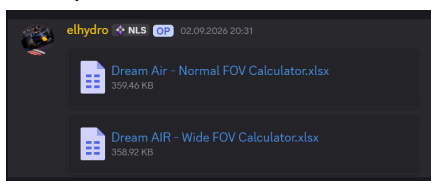


3. **Back up your files:** Rename the existing file `XR_API_LAYER_MBUCCHIA_toolkit.dll` in something like `XR_API_LAYER_MBUCCHIA_toolkit.back`.
4. **Copy the new DLL:** Drag and drop the file `XR_API_LAYER_MBUCCHIA_toolkit.dll` extracted from the PrimaShock ZIP file into the installation folder. If you are using the mod *3D BOOST + PERFORMANCE CROP*: also save the original file, then copy the new shader file (`postprocess.hlsl`) in `C:\Program Files\OpenXR-Toolkit\Shaders`.
5. **Update the shaders:** Open the folder `Shaders` in the installation directory.
6. **Save the original shaders:** Create a backup copy of the original shader files.
7. **Copy the new shaders:** Copy the contents of the folder `shader` taken from the mod's ZIP file in this directory.
8. **Check :** Open the OpenXR Toolkit companion application. It should now display "Primal Shock mod".

4. Choose the Excel file associated with your headset

<https://discordapp.com/channels/1371247671024680990/1530019814263754854>

example for the Pimax Dream Air



Open the Excel file and that's all for now (we'll need it later).

5. RESET TOOLKIT

To ensure a smooth start to the configuration, if you have already used the toolkit, I recommend resetting the toolkit (it's not mandatory but it's better in my opinion).

- Open the game
- open openxr toolkit
- Go to Menu+Reset+confirm
- **Close the toolkit, close the game, and completely restart the headset (restart both the headset and the service): that way we're starting from scratch.**

Next step: choosing the crop (The choice of crop = the number of pixels we are going to remove)

6. Crop selection

You now need to choose the crop settings: up (top), down (bottom), left/left, right/right, as well as left/right and right/left (those near the nose). Elhydro offers values depending on the car model; you can choose one of these and test it. These values are indicative, and you can choose the one that suits you best. Personal preference, visual preferences, GPU, CPU, and many other factors make it impossible to determine a single "best" value.

Target total vFOV (°)	Motorsport Visor Category (Example)	Up Target (°)	Down Target (°)	Total vFOV (°)
25°	Formula 1 / Open Wheel	12.50° (21.8%)	12.50° (21.8%)	25.00° (21.8%)
30°	Prototype / LMP	15.00° (26.2%)	15.00° (26.2%)	30.00° (26.2%)
35°	GTE / Le Mans Endurance	17.50° (30.5%)	17.50° (30.5%)	35.00° (30.5%)
40°	GT3 / GT4 Sports Cars	20.00° (34.9%)	20.00° (34.9%)	40.00° (34.9%)
★ 45°	Touring Cars / TCR	22.50° (39.2%)	22.50° (39.2%)	45.00° (39.2%)
50°	Rally / WRC / Off-Road	25.00° (43.6%)	25.00° (43.6%)	50.00° (43.6%)
55°	Track Day Street Cars	27.50° (47.9%)	27.50° (47.9%)	55.00° (47.9%)
60°	Open Visor / Max View	30.00° (52.3%)	30.00° (52.3%)	60.00° (52.3%)
★ Recommended	Balanced / All Around	22.50° (39.2%)	22.50° (39.2%)	45.00° (39.2%)

Personally, I use these values, but they again depend on my perception, the power of my computer, etc... (70 or 75 up and 90 or 95 down and 99% on the sides, the crop near the nose (L/R and R/L) is calculated via the Excel file)

Field of view	Simple Advanced
Up	70% (31.2°/31.2°)
Down	90% (-40.1°/-40.1°)
Left/Left	99% (-54.5°)
Left/Right	98% (44.6°)
Right/Left	98% (-44.6°)
Right/Right	99% (54.5°)

Percentages vs. Degrees: What You Need to Understand

- The percentage (%) simply indicates the portion of the original view that you retain. For example, 70% means that you keep 70% of the headset's native field of view and remove 30%.
- Degrees (°) measure the actual angular opening perceived by your eyes. It is this value in degrees that you must keep in mind, as it is the only universal optical unit:
 - Each helmet model has a different native field of view. An adjustment of 70% will not give the same angle on a Dream Air as on another helmet.
 - This is imperative for the following: in the ElHydro Excel table, you will have to enter the value in degrees (the number in parentheses in the Toolkit, e.g. 31.20°) and not the percentage, so that the solution formulas are accurate.

CHOOSE YOUR VALUES AND GO TO THE EXCEL FILE

7. VALUES IN EXCEL

Open the Excel file

		A	B	C	D	E	F	G	H	I	J	K	L	M	N			
6		User Input Fields (FOV Angles)						***Note: Super Sample per the table below leaving FOV values set						Pixel Per Degree Analysis				
8		Instructions:				FOV Orientation	FOV Values %## (Angle): USER INPUT = 3 VALUES ONLY UP - DOWN - LEFT-LEFT	Pimax Evo Scale - Corrected	Super Sampling	Width x Height	PPD Eff	PPD-Eff % diff						
9		- The values shown here should always match OpenXR Toolkit values. Every crop has a specific Resolution set.				Up	70%	31.20	0.89	80%	4540 2746	42	-14.29%					
10		- Only enter the FOV angles for Up / Down / Left-Left. The calculator determines all remaining values automatically.				Dn	90%	40.10	0.92	85%	4680 2831	43	-12.24%					
11		- From the Toolkit, enter the FOV angles shown in parentheses into the orange cells. - Example: 55% (28.10°) → enter 28.10°.				Left Left (LL)	99%	54.50	0.95	90%	4815 2913	44	-10.20%					
12		- Start LL at 100%. Decrease LL until you first notice the FOV getting smaller. Record the value immediately before the reduction.				DO NOT CHANGE LR / RL / RR - THEY ARE CALCULATED AUTOMATICALLY.						0.97	95%	4947 2992	46	-6.12%		
13		- Note: Image quality won't suffer from sharpening from values from 100% resolution up to the Ideal Value (PPD-Eff % diff). For performance purposes Start at 100%, and work your way up in resolution until FPS drops. (Don't let frame rate drop below 90 FPS.)				Left Right	99%	44.86	1.00	100%	5078 3078	47	-4.08%					
14		- Super Sample Yellow Highlighted Cell Should be the Target output for an Equivalent Pixel per Degree to the native resolution - this will change depending on the Size of the Crop				Right Left (RL)	99%	44.86	1.02	105%	5201 3146	48	-2.04%					
15						Right Right (RR)	99%	54.50	1.05	110%	5324 3220	49	0.00%					
16		Helmet Selection = Up + Down (angles) Cell EF24												1.07	115%	5443 3262	50	2.04%
17		Target total FOV (°)	Midpoint View Category (Example)	Up Target (°)	Down Target (°)	Total of FOV (°)	Legend:						1.10	120%	5560 3363	51	4.08%	
18							User Input						1.12	125%	5655 3433	52	6.12%	
19							Soft Fixed - can be edited						1.14	130%	5787 3503	53	8.16%	
20		25°	Frontside / 1 Open Helm	12.50° (50%)	12.50° (50%)	25.00° (100%)	RED FONT - Values of Interest						1.16	135%	5888 3567	54	10.20%	
21		30°	Frontside / 1 Open Helm	15.00° (50%)	15.00° (50%)	30.00° (100%)	Underlined RED FONT - Output Values						1.18	140%	6000 3633	55	12.24%	
22		35°	OTD / 1 Max Endurance	17.50° (50%)	17.50° (50%)	35.00° (100%)	Actual Helmet selection Degrees						1.20	145%	6112 3697	56	14.29%	
23		40°	OTD / 1 Max Endurance	20.00° (50%)	20.00° (50%)	40.00° (100%)	71.30						1.22	150%	6172 3760	57	16.33%	
24		45°	OTD / 1 Max Endurance	22.50° (50%)	22.50° (50%)	45.00° (100%)							1.24	155%	6319 3822	58	18.37%	
25		★ 45°	Twisting Cam / 7CR	22.50° (50%)	22.50° (50%)	45.00° (100%)							1.26	160%	6441 3883	59	20.41%	
26		50°	IRIS / WRC / Off Road	25.00° (50%)	25.00° (50%)	50.00° (100%)	EliHydro's Crop - Pimax Evo						1.28	165%	6520 3944	60	22.45%	
27		55°	IRIS / WRC / Off Road	27.50° (50%)	27.50° (50%)	55.00° (100%)	Ideal Values						1.30	170%	6616 4003	61	24.49%	
28		60°	Track Day Street Cam	30.00° (50%)	30.00° (50%)	60.00° (100%)	Top Center						1.32	175%	6715 4061	62	26.53%	
29		65°	Open View / Main View	32.50° (50%)	32.50° (50%)	65.00° (100%)	Bottom Center						1.34	180%	6810 4119	63	28.57%	
30		★ Recommended	Balanced / All Around	32.50° (50%)	32.50° (50%)	65.00° (100%)	L-EYE Left						1.36	185%	6904 4176	64	30.61%	
31							L-EYE Right						1.38	190%	6997 4232	65	32.65%	
32							R-EYE Left						1.40	195%	7088 4287	66	32.65%	
33							R-EYE Right						1.41	200%	7178 4342	67	34.69%	
34													1.43	205%	7268 4393	67	36.73%	
35													1.45	210%	7356 4449	68	38.78%	
36													1.47	215%	7443 4502	69	40.82%	
37													1.48	220%	7529 4554	70	42.86%	
38													1.50	225%	7614 4605	70	42.86%	

In green: The 3 squares in the green circle (Up, Dn, Left Left) :

These are the only values to be changed manually in the spreadsheet. You must enter the angles in degrees that you have chosen very precisely (here: 31.20°, 40.10° And 54.50°). All other values below (Left Right, Right Left, Right Right) are calculated automatically using the geometric formula: Never modify them by hand.

Be careful to enter the degrees and not the percentages!

In blue: the super sampling you choose

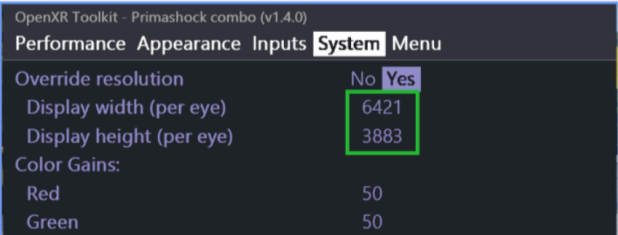
Line 110% SS (5324 × 3220) : This is the native clarity target (0.00% difference between PPD and Eff) By applying this step, you mathematically compensate for the loss of surface area and recover exactly the same pixel density per degree as the uncropped headset.

The higher tiers (Example: the 140% tier at 6006 × 3633) :

As soon as we cross the 0.00% line, we enter the suréchantillonnage pur (Super Sampling)

THIS IS AN EXAMPLE, THE VALUES ARE DIFFERENT DEPENDING ON THE HEADSET AND THE CROP, MAKE SURE YOU GO TO THE CORRECT EXCEL FILE AND LOOK AT THE CORRECT LINE ONCE YOU HAVE ENTERED THE 3 VALUES.
(The image shows the Pimax Dream Air only)

ONCE YOU HAVE ENTERED THE 3 VALUES, YOU NEED TO CHOOSE YOUR SUPER SAMPLING AND TRANSFER IT TO THE TOOLKIT
(for example, if I choose 160%, I will enter these values in the toolkit: 6421x3883)



1.00	100%	5076	3070	47	-4,08%
1,02	105%	5201	3146	48	-2,04%
1.05	110%	5324	3220	49	0,00%
1,07	115%	5443	3292	50	2,04%
1,10	120%	5560	3363	51	4,08%
1,12	125%	5675	3433	52	6,12%
1,14	130%	5787	3500	53	8,16%
1,16	135%	5898	3567	54	10,20%
1,18	140%	6006	3633	55	12,24%
1,20	145%	6112	3697	56	14,29%
1,22	150%	6217	3760	57	16,33%
1,24	155%	6319	3822	58	18,37%
1,26	160%	6421	3883	59	20,41%
1,28	165%	6520	3944	60	22,45%
1,30	170%	6618	4003	61	24,49%
1,32	175%	6715	4061	62	26,53%
1,34	180%	6810	4119	63	28,57%
1,36	185%	6904	4176	64	30,61%
1,38	190%	6997	4232	65	32,65%
1,40	195%	7088	4287	65	32,65%
1,41	200%	7178	4342	66	34,69%
1,43	205%	7268	4396	67	36,73%
1,45	210%	7356	4449	68	38,78%
1,47	215%	7443	4502	69	40,82%
1,48	220%	7529	4554	70	42,86%
1,50	225%	7614	4605	70	42,86%

To begin, try the two values highlighted in yellow that will appear in your Excel file. For example, start with the value 0.00% (in this case, 5324x3220) and increase it gradually according to your graphics card and PC.

In summary

- 1. Type the 3 values in the green box.
- 2. Choose your resolution
- 3. Remember the 6 crop numbers and the 2 resolution numbers

Up	70%	31,20
Dn	90%	40,10
Left Left (LL)	99%	54,50
DO NOT CHANGE LR / RL / RR — THEY ARE CALCULATED AUTOMATICALLY.		
Left Right	99%	44,86
Right Left (RL)	99%	44,86
Right Right (RR)	99%	54,50



0,97	95%	4947	2992	46	-6,12%
1.00	100%	5076	3070	47	-4,08%
1,02	105%	5201	3146	48	-2,04%
1.05	110%	5324	3220	49	0,00%
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1,10	120%	5560	3363	51	4,08%
1,12	125%	5675	3433	52	6,12%
1,14	130%	5787	3500	53	8,16%
1,16	135%	5898	3567	54	10,20%
1,18	140%	6006	3633	55	12,24%
1,20	145%	6112	3697	56	14,29%
1,22	150%	6217	3760	57	16,33%
1,24	155%	6319	3822	58	18,37%
1,26	160%	6421	3883	59	20,41%
1,28	165%	6520	3944	60	22,45%
1,30	170%	6618	4003	61	24,49%
1,32	175%	6715	4061	62	26,53%
1,34	180%	6810	4119	63	28,57%
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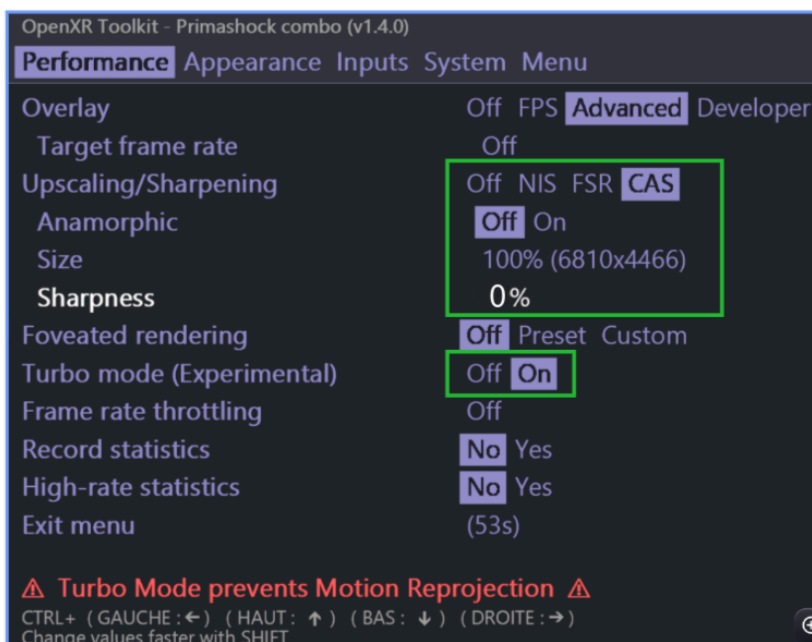
Next step: enter these values into the toolkit

(Don't worry, you can test and adjust your values to your liking directly in-game with the toolkit; you'll see the black bars when you apply a crop that's too aggressive, for example.)

Open the game and open the toolkit now

8. Configuration du toolkit

- Restart the game and the toolkit
- Underperformance: set CAS with Size to 100%. Adjust Sharpness according to your in-game results (test it in-game; I sometimes set it to 0%, sometimes 20%, and sometimes 75% depending on the game, so it's up to you).
- turn on turbo mode **ON**

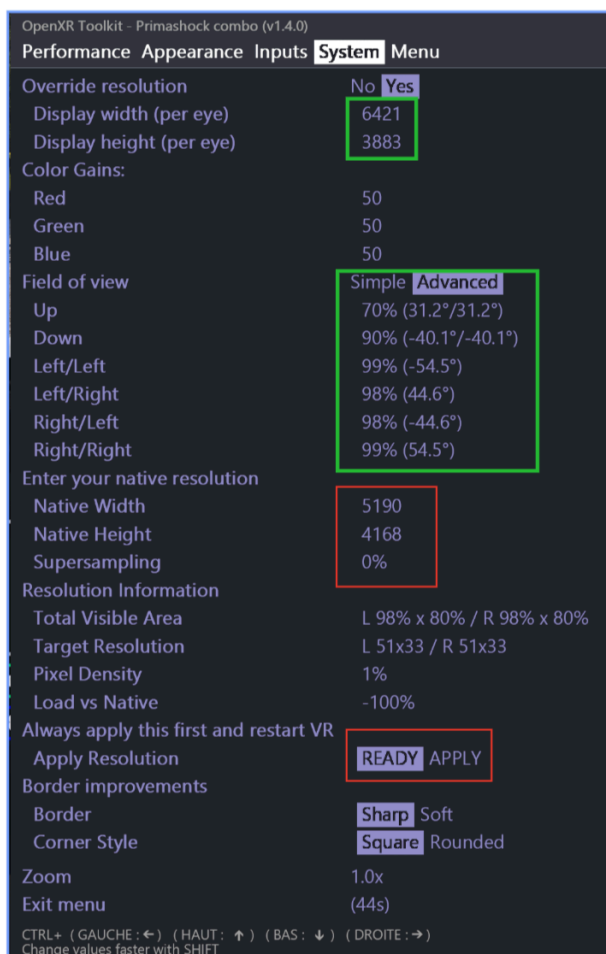


Next we move on to the part that interests us most, the CROP

CROP VALUES

- Go to “System”
- enter the 6 crop values
- , put **“override resolution” sur YES**
- Enter the two values in “override resolution” (width and height)
- Do not touch the values in the red box (regardless of the values entered), and do not click **NEVER** sur APPLY
- (If you accidentally clicked "apply" and confirmed, reset the toolkit and restart everything to be sure)

Reminder: Do not touch the things in the red boxes



- Close the toolkit by going to the “exit menu”
- Restart the game and it should be all good.

A few clarifications

- A. the **3d boost** Personally, I leave it at 0 (also recommended by elhydro)
- B. The drivers **SBOYS 3** must be disabled **absolutely** (The distortion profiles do not correspond well with the values in the Excel file.)
- C. In Pimax Play, the quality is set to 1.0.
- D. **Sun Glasses : ON (Light)**: An essential setting recommended by ElHydro on Dream Air OLED panels to remove the dull haze and open up the blacks; you should test it yourself. On some games, I prefer to disable it (in appearance, post-processing ON).
- E. **Vibration: 15 to 25**: To compensate for the flat colorimetry and enhance the contrast of the vibrators and the track (in appearance, post-processing ON)
- F. Cut the smart smoothing in Pimax Play
- G. **The magic anti-aliasing combo: NVIDIA MFAA + 2x MSAA in-game**
This is one of the secret pillars of the ElHydro method for obtaining an ultra-sharp image without compromising performance:
 - **In the NVIDIA Control Panel**: Go to *Manage 3D settings* > Select your game (or global settings) and adjust **Multi-Frame Sampled AA (MFAA)** on **Activated (On)**.
 - **In the game's graphics settings (e.g., AMS2)**: Set the anti-aliasing to **MSAA 2x (Low/Medium)** and you absolutely must disable the **Post-Processing Anti-Aliasing (Off)**.

Why this setting? NVIDIA's MFAA hardware technology combines time samples to automatically transform a 2x MSAA into a rendering equivalent to **MSAA 4x** while maintaining the negligible performance cost of 2x. Disabling post-processing AA avoids applying a destructive software blur filter.

H.

You need to test and play around with the crop size and super sampling (values in override resolution) in order to find the right balance in terms of fps in particular.

In any case, I'm giving you the process, but the values are often specific to each individual, to their own tastes, and depending on their computer, the results in terms of fps or quality may vary.

This mod was made for free by **Elhydro** To thank him, you can make a donation here, thank you in advance: <https://www.buymeacoffee.com/Elhydro>